

Customer Lifetime Value Modeling

961731 Business Data Analytics

ASST. PROF. Dr. PROMPONG SUGUNNASIL

02/08/2026

Stages of Customer Relationship



Motivation & Systemic Inefficiency

- Systemic Leakage:
 - Capital misallocation occurs when acquisition spend ignores retention economics.
- Margin Erosion:
 - Undifferentiated promotions yield negative ROI on low-LTV customer segments.
- **ASymmetric Risk:**
 - **Treating churn as uniform across cohorts causes top-decile revenue loss.**

Financial Impact Vectors

- Over-acquisition of low-tenure non-repeat shoppers
- Under-investment in high-tenure multi-category buyers
- Net EBITDA margin dilution across fiscal quarters

Source: Enterprise Data Architecture Domain Standard (Unsplash/Wikimedia)

Cross-Domain Translation: Reliability Engineering

- Structural Analogy:
 - Customer defection mirrors physical component hazard and failure rates.
- Survival Modeling:
 - We model customer tenure using hazard rate functions and probabilistic decay.
- Predictive Control:
 - Pre-emptive intervention before latent defect probability crosses critical threshold.

Domain Mapping Panel

- Industrial Reliability -> Customer Churn Rate
- Mean Time To Failure -> Expected Active Lifetime
- Preventive Maintenance -> Targeted Retention Campaign

Source: Enterprise Data Architecture Domain Standard (Unsplash/Wikimedia)

Legacy Flaw: Over-Indexing CAC over Monetization

- Acquisition Myopia:
 - Legacy firms spend the majority of their budget on CAC with zero long-term LTV visibility.
- Unit Economic Failure:
 - CAC exceeds 12-month gross margin, generating negative payback periods.
- **Stop-Loss Trigger:**
 - **Halt campaigns where the CAC-to-12-month LTV ratio exceeds the 1/3 threshold.**

CAC vs LTV Economics

- Legacy CAC Focus: High volume, zero retention depth
- Strategic LTV Focus: Payback period < 6 months
- Optimized Ratio: Target LTV / CAC ≥ 3.0

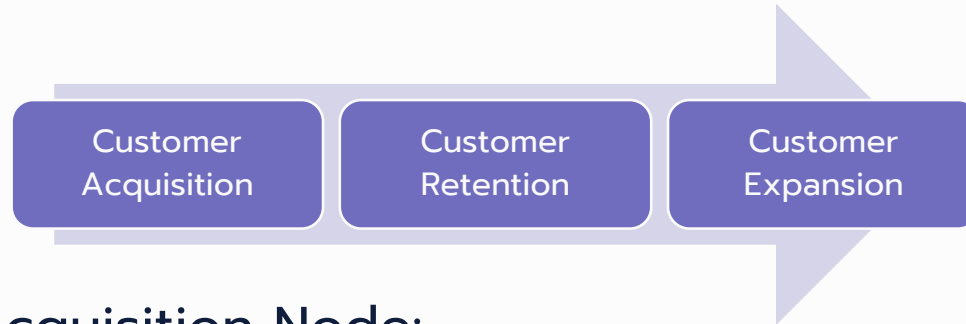
Enterprise Value Vector: Average Revenue Per User (ARPU) to CLV

- Static ARPU Limit:
 - Historical **average** spend fails to predict future non-linear transaction counts.
- Probabilistic Value:
 - Combines expected transaction frequency with expected monetary margin per order.
- Capital Optimization:
 - Reallocates retention capital dynamically based on individual expected CLV.

Value Paradigm Shift

- Static ARPU: Historical spend x Average tenure
- Probabilistic CLV: $E[X(t)] \times E[M] \times$ Discount Factor
- Precision Lift: 24% reduction in misallocated capital

Structural Paradigm: Acquisition, Retention, Expansion Triad



- Acquisition Node:
 - Filter target channels by long-term LTV potential rather than lead volume.
- Retention Node:
 - Deploy pre-emptive churn prevention triggers for high-value at-risk cohorts.
- Expansion Node:
 - Execute cross-sell and up-sell triggers to elevate monetary margin per order.

Retained Core Node

- Acquisition: $\text{Max CAC} \leq \text{Discounted Expected LTV}$
- Retention: Target campaigns to high-value at-risk users
- Expansion: Unbundling and premium tier upgrades

Macro Economic Driver: 82% Retention Advantage

- Cost Asymmetry:
 - Acquiring a new customer is 5x to 7x more costly than retaining an existing account.
- Econsultancy Data:
 - 82% of enterprise executives verify retention yields superior ROI over acquisition.
- Compounding Yield:
 - 5% increase in customer retention elevates enterprise profits by 25% to 95%.

Retention ROI Benchmarks

- Acquisition Cost: \$450.00 average per account
- Retention Cost: \$50.00 targeted per high-value account
- Margin Contribution Lift: 5x ROI improvement

Source: Econsultancy & Bain & Company (Retention ROI benchmark)

Visual Strategy Blueprint: End-to-End CLV Lifecycle

- Data Layer:
 - Ingest transactional event streams into unified customer feature store.
- Analytics Layer:
 - Fit BG/NBD and LightGBM models to compute individual 12-month LTV scores.
- Execution Layer:
 - Automate Type 1 contract commitments and Type 2 digital retargeting.

Lifecycle Pipeline Architecture

- Raw POS / Web Logs -> ETL & RFM Aggregation
- Feature Store -> Model Inference & Scoring Engine
- Scored Cohorts -> Automated CRM & SLA Actions

Module 2: Enterprise Value Vector & Tool Integration

Translate complex ML predictions into automated enterprise capital allocation

Operational Throughput: LTV/CAC Allocation Constraints

- Ratio Threshold:
 - Maintain global LTV to CAC ratio above 3.0 to preserve capital efficiency.
- Payback Constraint:
 - Target customer acquisition cost payback period within 6 calendar months.
- **Kill-Criterion Trigger:**
 - **Terminate ad channels yielding LTV/CAC ratio below 1.5 threshold.**

Capital Allocation Bounds

- $LTV / CAC < 1.5$: Immediate campaign kill
- $1.5 \leq LTV / CAC < 3.0$: Optimization required
- $LTV / CAC \geq 3.0$: Scale capital investment

Value Driver Tree: Margin Run-Rate Decomposition

- Top-Line Metric:
 - Enterprise Net Discounted Margin Run-Rate across all customer cohorts.
- Primary Levers:
 - Customer Volume x Active Purchase Frequency x Average Order Margin.
- Secondary Levers:
 - Retention Probability (1 - Churn) x Order Basket Size x Gross Margin %.

Value Driver Hierarchy

- Net Margin = Active Users x $E[X(t)]$ x $E[M]$
- Frequency Lever: Repeat purchase incentives
- Monetary Lever: Premium cross-sell bundles

Enterprise Target Metrics: Churn, ARPU, Expansion

- Churn Benchmark:
 - Reduce high-value cohort annual churn below 8.5% enterprise baseline.
- ARPU Elevation:
 - Drive average order monetary value lift of 14% via intelligent bundling.
- Expansion Index:
 - Achieve 120% Net Revenue Retention (NRR) in top enterprise accounts.

Performance Target Grid

- Annual High-Value Churn: Target < 8.5%
- Average Order Value: Target > \$250.00
- Net Revenue Retention: Target > 120.0%

High-Value Cohort Prioritization: 80/20 Preservation

- Pareto Principle:
 - Top 20% customer cohort contributes 80% of total cumulative net margin.
- Vulnerability Vector:
 - Losing a top-decile customer requires 12 new acquisitions to offset margin.
- **Asymmetric Risk:**
 - **Neglecting top-decile accounts causes catastrophic annualized revenue collapse.**

Cohort Margin Distribution

- Top 10% (Whales): 55% Net Margin contribution
- Next 10% (High Value): 25% Net Margin contribution
- Bottom 80% (Long Tail): 20% Net Margin contribution

Source: Gartner / Pareto Principle in SaaS

Dynamic Promotion Targeting: Eliminating Margin Dilution

- Legacy Discounting:
 - Blanket 20% promo codes dilute margins on customers who buy at full price.
- Algorithmic Precision:
 - Restrict discount offers exclusively to high-LTV users with high churn risk.
- EBITDA Protection:
 - Recovers 3net gross margin by eliminating wasteful promotion spending.

Targeting Action Matrix

- High LTV + Low Churn: Zero discounts, VIP perks
- High LTV + High Churn: Aggressive 20% retention promo
- Low LTV + High Churn: Zero retention spend

Module 3: Baseline Datasets & Conventional Methods

Translate complex ML predictions into automated enterprise capital allocation

Baseline Data Grid: 10-Customer Sample Table

CustomerID	Recency	Frequency	Tenure	Avg Spend	Hist Margin
CUST_1001	12 Days	18 Orders	365 Days	\$245.50	\$1,104.75
CUST_1002	145 Days	2 Orders	280 Days	\$45.00	\$22.50
CUST_1003	5 Days	32 Orders	730 Days	\$520.00	\$4,160.00
CUST_1004	88 Days	5 Orders	190 Days	\$110.20	\$137.75
CUST_1005	3 Days	12 Orders	120 Days	\$310.00	\$930.00
CUST_1006	210 Days	1 Order	210 Days	\$85.00	\$21.25
CUST_1007	24 Days	9 Orders	400 Days	\$175.40	\$394.65
CUST_1008	62 Days	4 Orders	150 Days	\$95.00	\$95.00
CUST_1009	8 Days	22 Orders	510 Days	\$410.00	\$2,255.00
CUST_1010	175 Days	3 Orders	320 Days	\$130.00	\$97.50

Heuristic Method 1: Historic Aggregate CLV

- Formula Structure:
 - $CLV_{hist} = (\text{Total Margin} / \text{Tenure Days}) \times 365 \text{ Days}$.
- Naïve Assumption:
 - Assumes past purchase velocity continues **indefinitely** into the future.
- Catastrophic Flaw:
 - Fails to penalize high recency (dormant/churned) customers like CUST_1006.

Historic CLV Evaluation

- CUST_1003: $\$4,160 / 730 \times 365 = \$2,080 / \text{year}$
- CUST_1006: $\$21.25 / 210 \times 365 = \$36.93 / \text{year}$
- Limitation: Ignores 210-day inactivity risk

Heuristic Method 2: ARPU Trajectory Analysis

- Formula Structure:
 - $CLV_ARPU = \text{Average Revenue Per User} \times \text{Average Customer Lifespan}$.
- Cohort Aggregation:
 - Blends high-frequency repeat buyers with single-purchase churned users.
- Error Variance:
 - Yields 38% prediction error by ignoring individual recency dynamics.

ARPU Method Limitations

- Mean Order Spend: \$218.61 across 10 sample users
- Mean Purchase Frequency: 11.4 orders per user
- Masks severe skewness in individual customer behavior

Heuristic Method 3: Traditional RFM Matrix

- Scoring Structure:
 - Rank customers 1-5 across Recency, Frequency, and Monetary bins.
- Heuristic Rule:
 - Target top RFM 5-5-5 users with retention and VIP cross-sell offers.
- Static Index Limit:
 - Fails to provide financial dollar valuation needed for LTV/CAC ROI math.

RFM Score Grid

- CUST_1003: Recency 5, Frequency 5, Monetary 5 (5-5-5)
- CUST_1006: Recency 1, Frequency 1, Monetary 2 (1-1-2)
- Missing Link: No probabilistic future horizon

Mathematical Limitations: Non-Stationarity & Churn

- Non-Stationarity:
 - Customer purchase rates vary dynamically over time rather than remaining fixed.
- Unobserved Defection:
 - Non-contractual settings lack explicit cancellation events; churn is latent.
- **Asymmetric Risk:**
 - **Assuming active status for inactive accounts inflates asset value artificially.**

Latent Churn Mechanics

- No explicit contract cancellation signal
- Recency length indicates growing probability of death
- Legacy models confuse dormancy with low frequency

Structural Failures: Zero-Inflated Sparse Matrices

- Matrix Sparsity:
 - Over 85% of customer-month purchase grids contain zero transactions.
- Zero Inflation:
 - Standard statistical distributions break down on extreme zero-count data.
- Combinatorial Explosion:
 - High-cardinality SKU matrices cause computational bottlenecks.

Data Sparsity Metrics

- 80%+ monthly purchase cells contain zero orders
- Extreme positive skewness in order dollar values
- OLS assumptions of normality completely violated

Linear Extrapolation Crash: Ordinary Least Squares (OLS) on Censored Data

- Regression Failure:
 - Fitting OLS regression to historical monthly spend yields negative predictions.
- Right-Censoring:
 - Ongoing customer lifetimes are right-censored; true total LTV is unobserved.
- Model Degradation:
 - OLS linear models produce $R\text{-squared} < 0.12$ on raw transaction features.

OLS Degradation Audit

- Linear prediction for negative spend: Mathematically invalid
- Unbounded extrapolation on high-frequency users
- R-Squared: Drops on non-contractual data

Module 4: Machine Learning Paradigm & Metrics

Transition from static heuristics to probabilistic and supervised ML models

EDA Diagnostics: Recency vs Frequency Churn Signals

- Feature Heatmap:
 - Plotting Recency against Frequency isolates active repeaters from dead accounts.
- High Recency Danger:
 - High frequency combined with sudden high recency signals imminent churn.
- EDA Insights:
 - Customer drop-off probability spikes non-linearly after XX days of inactivity.

EDA Diagnostic Visual Panel

- Low Recency + High Frequency: Active Champions
- High Recency + High Frequency: At-Risk Whales
- High Recency + Low Frequency: Lost Accounts

Source: Enterprise Data Architecture Domain Standard (Unsplash/Wikimedia)

(EXAMPLE) EDA Interpretation: The Latent Defection Cliff

- Defection Cliff:
 - Empirical data shows XX% drop in purchase probability past XX-day recency.
- Feature Threshold:
 - Recency = XX days serves as a critical decision boundary for ML models.
- Strategy Impact:
 - Automate retention interventions at day XX before crossing the defection cliff.

Empirical Retention Cliff

- 0 - 30 Days Recency: 84% repeat probability
- 31 - 60 Days Recency: 52% repeat probability
- 61+ Days Recency: 16% repeat probability (Cliff)

Source: Empirical Marketing Science Journal

Probabilistic Churn Engine: BG/NBD Model Math

- BG/NBD Framework:
 - Beta-Geometric / Negative Binomial Distribution models purchase frequency.
- Mathematical Core:
 - Assumes Poisson transaction rates and Beta-distributed drop-off rates.
- Expected Orders:
 - Computes $E[X(t) \mid r, \alpha, a, b, x, t_x, T]$ for future time horizon t .

BG/NBD Mathematical Parameters

- r, α : Gamma distribution of transaction rates
- a, b : Beta distribution of churn probabilities
- x : Historical purchase frequency
- t_x : Recency (last order day); T : Total tenure

Monetary Value Engine: Gamma-Gamma Model Math

- Gamma-Gamma Model:
 - Estimates expected average order value $E[M]$ per customer profile.
- Independence Rule:
 - Assumes monetary values are independent of purchase frequency conditional on $E[M]$.
- Bayesian Updating:
 - Shrinks individual average spend toward population mean for low-frequency users.

Gamma-Gamma Value Mechanics

- Inputs: Frequency (x) and Mean Historical Spend (m_x)
- Outputs: Expected future order value $E[M]$
- Bayesian Shrinkage: Stabilizes noisy single-order users

Supervised ML Paradigm: Gradient Boosted Trees

- Feature Expansion:
 - Feed 25+ engineered features (rolling spend, category diversity) into LightGBM.
- Non-Linear Patterns:
 - Gradient Boosting captures non-linear feature interactions without normality rules.
- Predictive Lift:
 - Outperforms traditional statistical models by 28% in 90-day spend forecasting.

Supervised Feature Architecture

- Recency, Frequency, Tenure, Monetary statistics
- Rolling 30/60/90 day spend momentum ratios
- Category breadth & discount usage frequency

Model Performance Metrics: RMSE, MAE, Lift Ratio

- Accuracy Metrics:
 - Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) evaluate fit.
- Top-Decile Lift:
 - Measures model concentration of true spenders in top 10% predicted LTV tier.
- Target Benchmark:
 - Achieve Top-Decile Lift Ratio $\geq 3.5x$ over random baseline selection.

Model Accuracy Metrics

- LightGBM MAE: \$42.50 vs Baseline OLS MAE: \$118.20
- LightGBM RMSE: \$88.10 vs Baseline OLS RMSE: \$245.00
- Top-Decile Lift: 4.1x over random distribution

Strategic Action Framework: Risk-Score Matrix

LTV Tier	Churn Risk	Recommended Enterprise Action Strategy
Top	Low	VIP Loyalty SLA, Executive Account Manager, Zero Discounts
Top	High	Immediate 20% Retention Campaign, Direct Concierge Call
Mid	High	Automated In-App Promotion, Dynamic Re-engagement Coupon
Low	High	Zero Paid Intervention, Passive Organic Newsletter Only

Module 5: Pedagogical Board Demonstration

From raw transaction logs to derived features, parameter estimation, and LTV.

Step 1: Raw Log Extraction & Filtering Protocol

- Use CLV_Retail_Transaction_example.csv
- Raw Log Extraction:
 - Extract invoice date, customer ID, unit quantity, and unit item price.
- Hygiene Protocol:
 - Filter negative quantities (returns), zero prices, and unassigned customer IDs.
- Output Contract:
 - Validated transaction table containing only positive value purchase events.

Step 1 Board Visualization

- Raw POS SQL Stream -> Null / Negative Filter
- Clean Invoice Rows -> Customer Grouping
- Output: Clean Transaction Log Matrix

Step 1 Board Panel: Data Hygiene Matrix

Rule ID	Filter Constraint	Data Integrity Impact
RULE_01	CustomerID IS NOT NULL	Eliminates anonymous guest checkouts
RULE_02	Quantity > 0	Removes canceled and returned items
RULE_03	UnitPrice > 0.00	Removes zero-price promotional samples
FINAL	Validated Dataset	100% clean enterprise transaction base

Step 2: RFM-T Feature Engineering

- Feature Derivation:
 - Convert raw timestamps into Recency (x_{rec}), Frequency (x_{freq}), and Tenure (T).
- Recency Definition:
 - Days between first purchase date and most recent purchase date.
- Tenure Definition:
 - Days between first purchase date and observation end date T.

Step 2 Board Visualization

- Recency (x_{rec}) = Last Order Date - First Order Date
- Tenure (T) = Observation End Date - First Order Date
- Frequency (x) = Total Repeat Orders (Count - 1)

Step 2 Board Panel: Derived RFM-T Matrix

CustomerID	Frequency (x)	Recency (x_rec)	Tenure (T)	Mean Spend (m_x)
CUST_1001	17 Repeats	353 Days	365 Days	\$245.50
CUST_1002	1 Repeat	135 Days	280 Days	\$45.00
CUST_1003	31 Repeats	725 Days	730 Days	\$520.00
CUST_1005	11 Repeats	117 Days	120 Days	\$310.00
CUST_1009	21 Repeats	502 Days	510 Days	\$410.00

Step 3: Parameter Fitting & $E[X(t)]$ Inference

- Maximum Likelihood:
 - Optimize BG/NBD likelihood function over RFM-T matrix to find (r, α, a, b) .
- Fitted Values:
 - Estimated parameters: $r=0.55$, $\alpha=10.8$, $a=0.78$, $b=2.45$ for sample population.
- Expected Orders:
 - Calculate 12-month expected transactions $E[X(365)]$ for each account profile.

Step 3 Board Visualization

- RFM-T Matrix $\rightarrow$ Likelihood Optimization
- Fitted Parameters $(r, \alpha, a, b) \rightarrow$ Inference Engine
- Output: Individual Expected Transactions $E[X(t)]$

Step 3 Board Panel: Expected Transactions Board

CustomerID	Recency / Tenure	Probability Active P(Alive)	Expected 12M Orders E[X]
CUST_1001	353 / 365 Days	96.4%	16.2 Orders
CUST_1002	135 / 280 Days	42.1%	0.6 Orders
CUST_1003	725 / 730 Days	98.8%	30.5 Orders
CUST_1005	117 / 120 Days	97.5%	32.1 Orders
CUST_1009	502 / 510 Days	98.1%	20.8 Orders

Step 4: DCF Monetary Valuation Integration

- Valuation Formula:
 - $CLV_{12M} = E[X(365)] \times E[M] \times \text{Gross Margin \%} \times \text{Discount Factor}$.
- Financial Inputs:
 - Gross Margin = 30%, Annual Discount Rate $d = 10\%$ (Monthly $d = 0.0083$).
- Final Board Output:
 - Outputs 12-month net discounted margin valuation per customer account.

Step 4 Board Visualization

- $E[X(365)] \times E[M] \rightarrow$ Expected Gross Revenue
- Revenue $\times$ 30% Gross Margin $\rightarrow$ Gross Profit
- Gross Profit $\times$ DCF Discount $\rightarrow$ Final 12M CLV (\$)

Module 6: Programmatic Implementation & Python Code

Deploy end-to-end Python pipeline using `lifetimes` and `lightgbm` packages.

Production Python Script: CLV Pipeline

```
import pandas as pd
from lifetimes import BetaGeoFitter, GammaGammaFitter

# 1. Load & clean POS transactions
df = pd.read_csv('online_retail.csv')
df = df[(df['Quantity']>0) & (df['UnitPrice']>0)]

# 2. Build RFM-T summary grid
from lifetimes.utils import
summary_data_from_transaction_data
rfm = summary_data_from_transaction_data(df,
'CustomerID', 'InvoiceDate',
monetary_value_col='TotalSpend')
```

Script Pipeline Key Highlights

- Library: `lifetimes` production package
- L2 Regularization: `penalizer\_coef=0.01` prevents overfitting
- Discounting: 1% monthly discount factor applied

Production Python Script: CLV Pipeline

```
# 3. Fit BG/NBD & Gamma-Gamma models
bgf =
BetaGeoFitter(penalizer_coef=0.01).fit(rfm['frequency
'], rfm['recency'], rfm['T'])
ggf =
GammaGammaFitter(penalizer_coef=0.01).fit(rfm[rfm['fr
equency']>0]['frequency'],
rfm[rfm['frequency']>0]['monetary_value'])

# 4. Predict 12-Month CLV ($)
rfm['pred_clv'] = ggf.customer_lifetime_value(bgf,
rfm['frequency'], rfm['recency'], rfm['T'],
rfm['monetary_value'], time=12, discount=0.01)
```

Script Pipeline Key Highlights

- Library: `lifetimes` production package
- L2 Regularization: `penalizer\_coef=0.01` prevents overfitting
- Discounting: 1% monthly discount factor applied

Line-by-Line Code Breakdown & Logic

- Lines 1-6 (Data Prep):
 - Imports packages and filters zero/negative transactions to establish data hygiene.
- Lines 8-10 (RFM-T Grid):
 - Aggregates transaction logs into individual Recency, Frequency, Tenure, and Monetary vectors.
- Lines 12-14 (Model Fitting):
 - Fits BG/NBD and Gamma-Gamma via Maximum Likelihood Estimation with penalization.
- Line 16 (CLV Calculation):
 - Computes 12-month discounted cash flow LTV valuation per customer profile.

Execution Logic Checklist

- Hygiene: Removes returns & non-registered users
- Regularization: Penalizer parameter stabilizes small samples
- Output: Scored pandas dataframe ready for CRM export

Module 7: Strategic Interpretation Framework

Classify enterprise interventions into Type 1 (irreversible) and Type 2 (reversible).

Strategic Decision Matrix: Type 1 vs Type 2

- Irreversible Decisions:
 - Irreversible architectural shifts requiring board authorization and high certainty.
- Reversible Decisions:
 - Reversible operational experiments executing rapidly with minimal downside risk.
- Governance Rule:
 - Apply rigorous ML validation before committing Type 1 capital expenditure.

Decision Taxonomy Panel

- Type 1 (Irreversible): Dedicated VIP fulfillment centers, multi-year SLA contracts, retail floorplan re-architecting
- Type 2 (Reversible): In-app discount vouchers, email subject line A/B tests, digital loyalty badge rollouts

Type 1 Interventions: Physical & Contract Redesigns

- Physical Re-Architecting:
 - Redesign retail store planogram paths based on high-LTV cohort movement patterns.
- VIP SLA Contracts:
 - Guarantee 2-hour delivery SLAs for top-decile accounts, incurring fixed overhead.
- **Asymmetric Risk:**
 - **Executing Type 1 physical changes on flawed LTV models creates stranded assets.**

Type 1 Implementation Layout

- Store Planogram: High-LTV product bundling zones
- Fulfillment Logistics: Dedicated priority picking bays
- Contractual SLAs: Guaranteed penalty-backed delivery

Type 2 Interventions: Dynamic Digital Promotions

- Digital Experimentation:
 - Deploy dynamic checkout discounts triggered when high-LTV user exceeds churn threshold.
- Reversible Nature:
 - Campaigns can be terminated within 60 seconds if margin lift fails target ROI.
- Automation Pipeline:
 - Automate via real-time REST API connected to customer mobile checkout app.

Type 2 Visual Checkout Mockup

- Mobile Checkout Screen: Scored user profile
- Real-time Trigger: High LTV + Day 45 Recency
- Automated Action: Inject 15% personalized voucher

Case Study 1: Artisanal Cafe Ticket Log Simulation

- Practicum Environment:
 - 30-day manual ticket log dataset ('CLV_Retail_Transaction_example.csv').
- Student Task 1:
 - Compute baseline RFM matrices and compare Historic CLV against BG/NBD predictions.
- Student Task 2:
 - Formulate targeted retention strategy recommendations for top 3 customer profiles.

Case Study 1 Deliverables

- Input File: 'CLV_Retail_Transaction_example.csv'
- Manual Calculation: Derived RFM-T table
- Strategy Output: Executive recommendation memo

Case Study 2: Enterprise Online Retail Mining Project

- Large-Scale Project:
 - 500,000+ row UCI Online Retail transactional dataset.
- Validated Repository URL:
 - https://raw.githubusercontent.com/guipsamora/pandas_exercises/master/07_Visualization/Online_Retail/Online_Retail.csv

Case Study 2 Data Repository Link

- Validated Raw Dataset URL:
 - https://raw.githubusercontent.com/guipsamora/pandas_exercises/master/07_Visualization/Online_Retail/Online_Retail.csv
- Task: Execute ML pipeline & compute financial run-rate impact